



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Experiment 21. Syntax-Driven Semantic Analysis (Noun Phrase Extraction)

Program

```
import nltk

nltk.download('punkt')

nltk.download('averaged_perceptron_tagger')

nltk.download('maxent_ne_chunker')

nltk.download('words')

sentence = input("Enter sentence: ")

tokens = nltk.word_tokenize(sentence)

tags = nltk.pos_tag(tokens)

grammar = "NP: {<DT>?<JJ>*<NN.*>+}"

cp = nltk.RegexpParser(grammar)

tree = cp.parse(tags)

print("Noun Phrases:")

for subtree in tree.subtrees():

    if subtree.label() == "NP":

        phrase = " ".join(word for word, tag in subtree.leaves())

        print(phrase)
```

Output

```
[4] ✓ 9s
▶ for subtree in tree.subtrees():
    if subtree.label() == "NP":
        phrase = " ".join(word for word, tag in subtree.leaves())
        print(phrase)

... [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger is already up-to-
[nltk_data] date!
[nltk_data] Downloading package maxent_ne_chunker to
[nltk_data] /root/nltk_data...
[nltk_data] Package maxent_ne_chunker is already up-to-date!
[nltk_data] Downloading package words to /root/nltk_data...
[nltk_data] Package words is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data] /root/nltk_data...
[nltk_data] Unzipping taggers/averaged_perceptron_tagger_eng.zip.
Enter sentence: The smart student solved the difficult problem.
Noun Phrases:
The smart student
the difficult problem
```

Google Colab is available in VS Code!

